Exploratory Data Analysis: Iris Measurements

Project: Iris dataset exploration
Focus: Data understanding, visual analysis, and evidence-based conclusions
Tools: Python, pandas, Matplotlib, and Seaborn

Research question

How clearly do four basic flower measurements distinguish the three Iris species in this dataset, and which relationships are worth investigating further?

This notebook explores the data through descriptive statistics and visualisation. It is an exploratory data analysis (EDA).

Analysis outline

1. Understand the dataset and check its quality.
2. Compare the distributions of the four measurements.
3. Explore a small set of derived measurements.
4. Examine correlations and relationships between variables.
5. Summarise the findings, caveats, and possible next steps.

1. Dataset overview

The dataset contains measurements of three Iris species: *Iris-setosa*, *Iris-versicolor*, and *Iris-virginica*.

Each row represents one flower. Four measurements are recorded in centimetres:

- Sepal length
- Sepal width
- Petal length
- Petal width

The `species` column identifies the species recorded for each flower.

Later in this analysis, we calculate a few additional values from the original measurements. These provide alternative ways to explore the data, but they are not direct measurements of botanical surface area.

iris-setosa



iris-versicolor



iris-virginica



```
Source file: 25__iris.csv
Dataset dimensions: 150 rows × 5 columns
```

Dataset at a glance:

Rows: **150** | Numeric measurements: **4** | Species: **3**

Species labels: **Iris-setosa, Iris-versicolor, Iris-virginica**

Example records:

sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
6.10	2.80	4.70	1.20	Iris-versicolor
5.70	3.80	1.70	0.30	Iris-setosa
7.70	2.60	6.90	2.30	Iris-virginica
6.00	2.90	4.50	1.50	Iris-versicolor
6.80	2.80	4.80	1.40	Iris-versicolor

2. Data structure and quality

Before comparing the species, we checked the dataset for missing values and identical records.

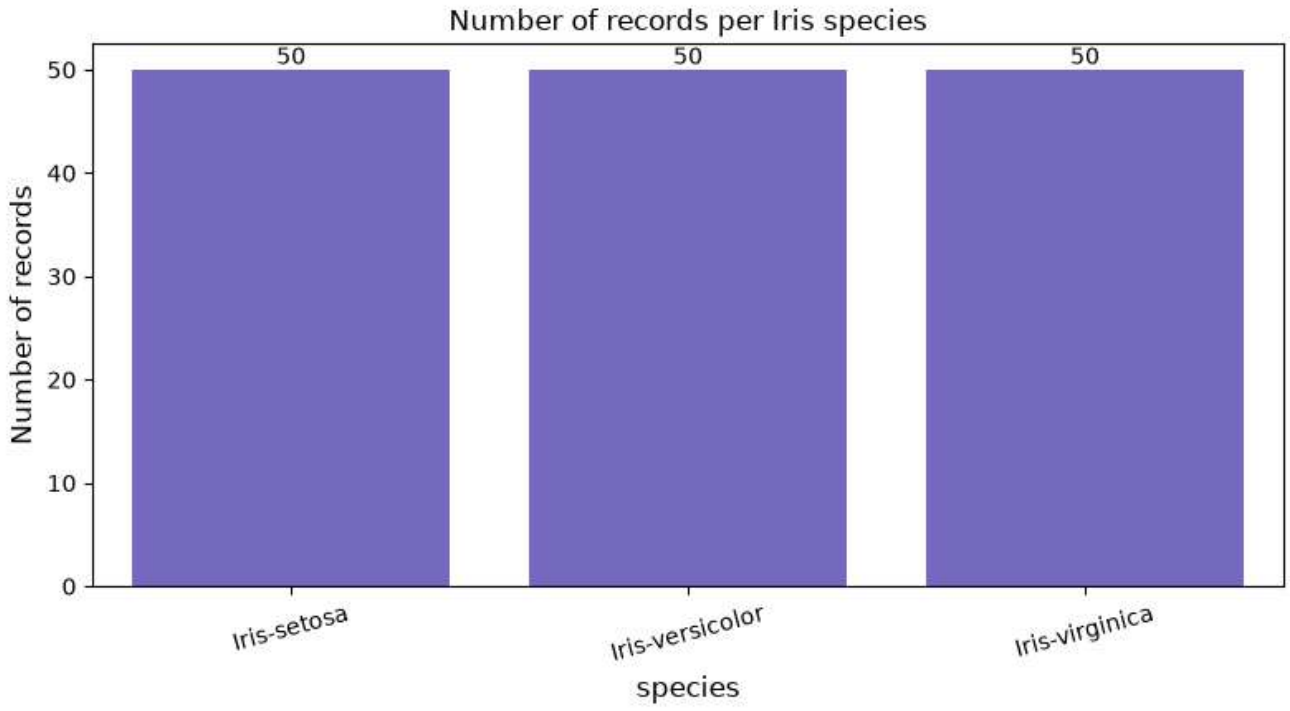
No missing values were found. Three rows are exact duplicates. They have been retained because identical measurements alone do not prove that a record is incorrect.

	missing values	missing (%)	unique values	missing data - status summary
sepal length (cm)	0	0.00%	35	**OK**
sepal width (cm)	0	0.00%	23	**OK**
petal length (cm)	0	0.00%	43	**OK**
petal width (cm)	0	0.00%	22	**OK**
species	0	0.00%	3	**OK**

```
Exact duplicate rows: 3
```

Class balance

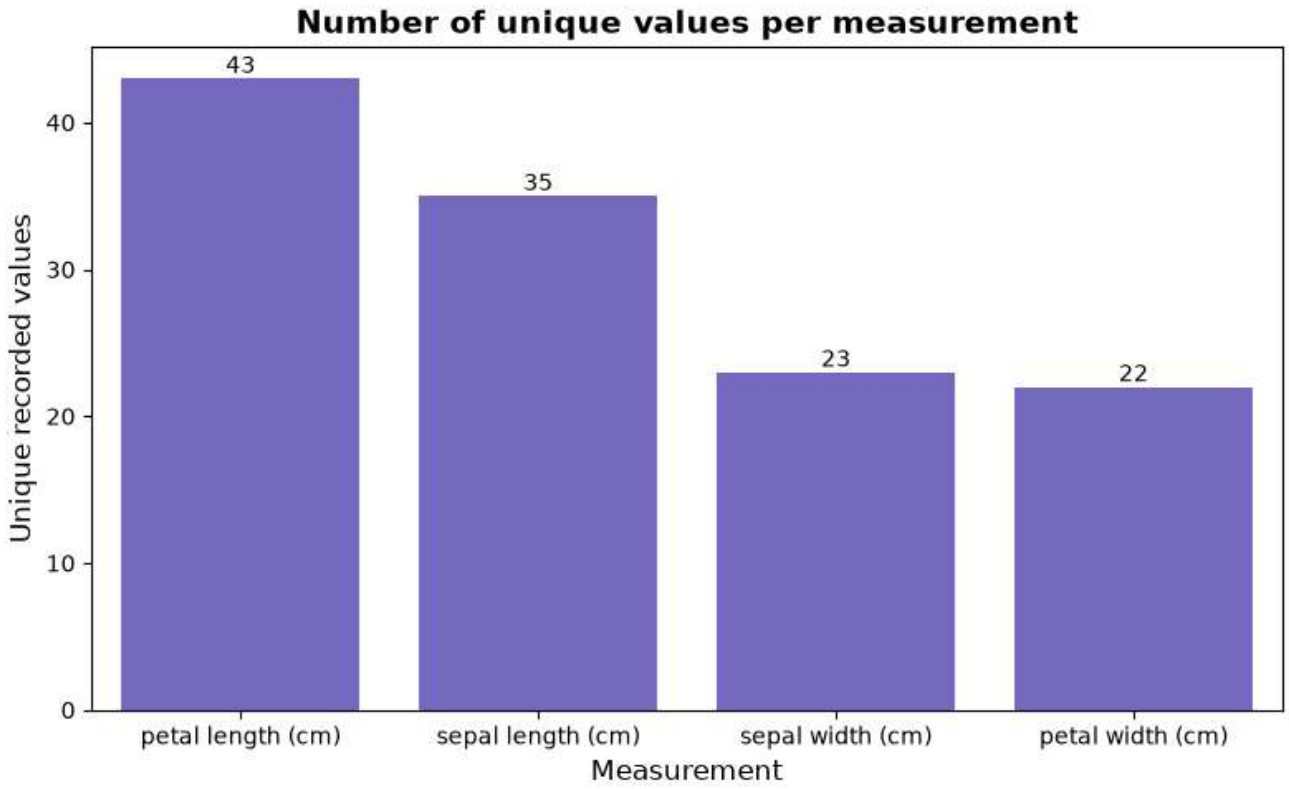
The three species have **50 records each**, so the dataset is balanced by class. This makes the descriptive comparisons easier to interpret, although class balance alone does not prove that the measurements distinguish the species.



3. Exploring individual measurements

This section looks at each measurement separately. The charts and summary table help us understand the range of values, how widely they vary, and how their distributions differ.

A measurement with more distinct recorded values is not necessarily more useful for distinguishing species. To understand which measurements show the clearest differences, we need to compare their distributions across the three species.



What does this chart show?

The chart shows how many distinct values were recorded for each measurement. It helps us understand the variety of recorded values, but it does not tell us which measurement best distinguishes the species.

Descriptive statistics

The first table summarises the four measurements across all 150 flowers. The tables that follow show the same measurements separately for each species, making differences between species easier to compare.

	count	mean	std	min	25%	50%	75%	max
sepal length (cm)	150.00	5.84	0.83	4.30	5.10	5.80	6.40	7.90
sepal width (cm)	150.00	3.05	0.43	2.00	2.80	3.00	3.30	4.40
petal length (cm)	150.00	3.76	1.76	1.00	1.60	4.35	5.10	6.90
petal width (cm)	150.00	1.20	0.76	0.10	0.30	1.30	1.80	2.50

Comparing measurements across species

When all flowers are considered together, petal length varies widely. This is partly because the three species have different typical petal sizes.

Looking at each species separately makes it easier to see how similar or varied the measurements are within each group. The tables below help us compare these differences.

species		sepal length (cm)						
		count	mean	std	min	25%	50%	75%	max
0	Iris-setosa	50.000	5.006	0.352	4.300	4.800	5.000	5.200	5.800
1	Iris-versicolor	50.000	5.936	0.516	4.900	5.600	5.900	6.300	7.000
2	Iris-virginica	50.000	6.588	0.636	4.900	6.225	6.500	6.900	7.900

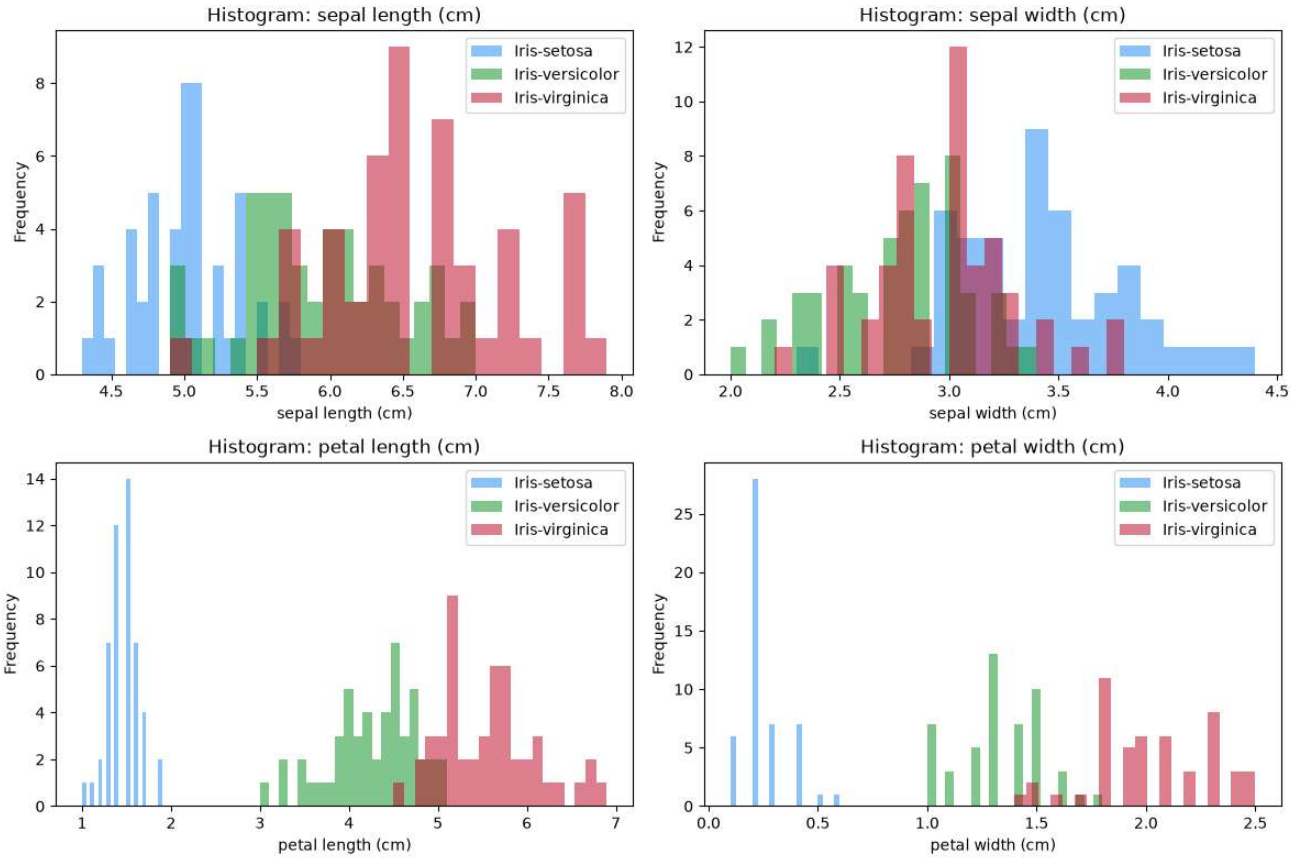
species		sepal width (cm)						
		count	mean	std	min	25%	50%	75%	max
0	Iris-setosa	50.000	3.418	0.381	2.300	3.125	3.400	3.675	4.400
1	Iris-versicolor	50.000	2.770	0.314	2.000	2.525	2.800	3.000	3.400
2	Iris-virginica	50.000	2.974	0.322	2.200	2.800	3.000	3.175	3.800

species		petal length (cm)						
		count	mean	std	min	25%	50%	75%	max
0	Iris-setosa	50.000	1.464	0.174	1.000	1.400	1.500	1.575	1.900
1	Iris-versicolor	50.000	4.260	0.470	3.000	4.000	4.350	4.600	5.100
2	Iris-virginica	50.000	5.552	0.552	4.500	5.100	5.550	5.875	6.900

species		petal width (cm)						
		count	mean	std	min	25%	50%	75%	max
0	Iris-setosa	50.000	0.244	0.107	0.100	0.200	0.200	0.300	0.600
1	Iris-versicolor	50.000	1.326	0.198	1.000	1.200	1.300	1.500	1.800
2	Iris-virginica	50.000	2.026	0.275	1.400	1.800	2.000	2.300	2.500

Histograms by species

Each panel shows the distribution of one measurement, with colour indicating species. Notice where the distributions separate and where they overlap. Overlap means that this single measurement does not uniquely identify every flower in the overlapping range.



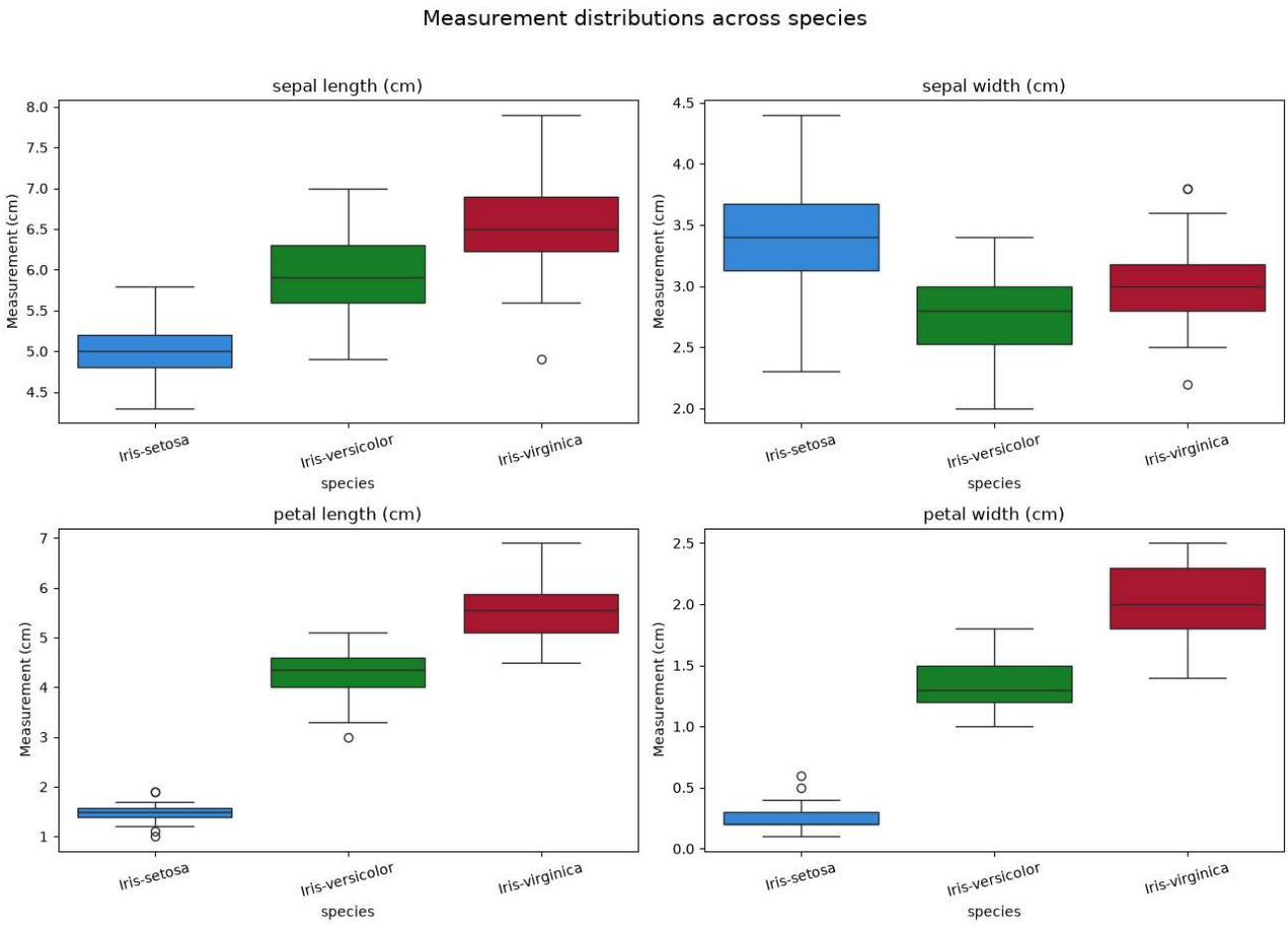
Histogram interpretation

The petal-length and petal-width distributions show a clear separation for *Iris-setosa*. The distributions for *Iris-versicolor* and *Iris-virginica* overlap more, while sepal-based distributions overlap across species more broadly. These are visual observations, not error rates or classification results.

4. Comparing measurements across species

These charts compare the typical values and spread of each measurement across the three species.

The line inside each box marks the median, while the box shows the middle half of the observations. Points beyond the whiskers represent unusually distant values, but they are not automatically errors.



Interpretation of the boxplots

Compare the separation between the species' medians with the overlap between their boxes and whiskers. The petal measurements show clearer differences between groups than the sepal measurements, especially for *Iris-setosa*. *Iris-versicolor* and *Iris-virginica* still overlap for some measurements, so these plots do not justify claiming perfect separation.

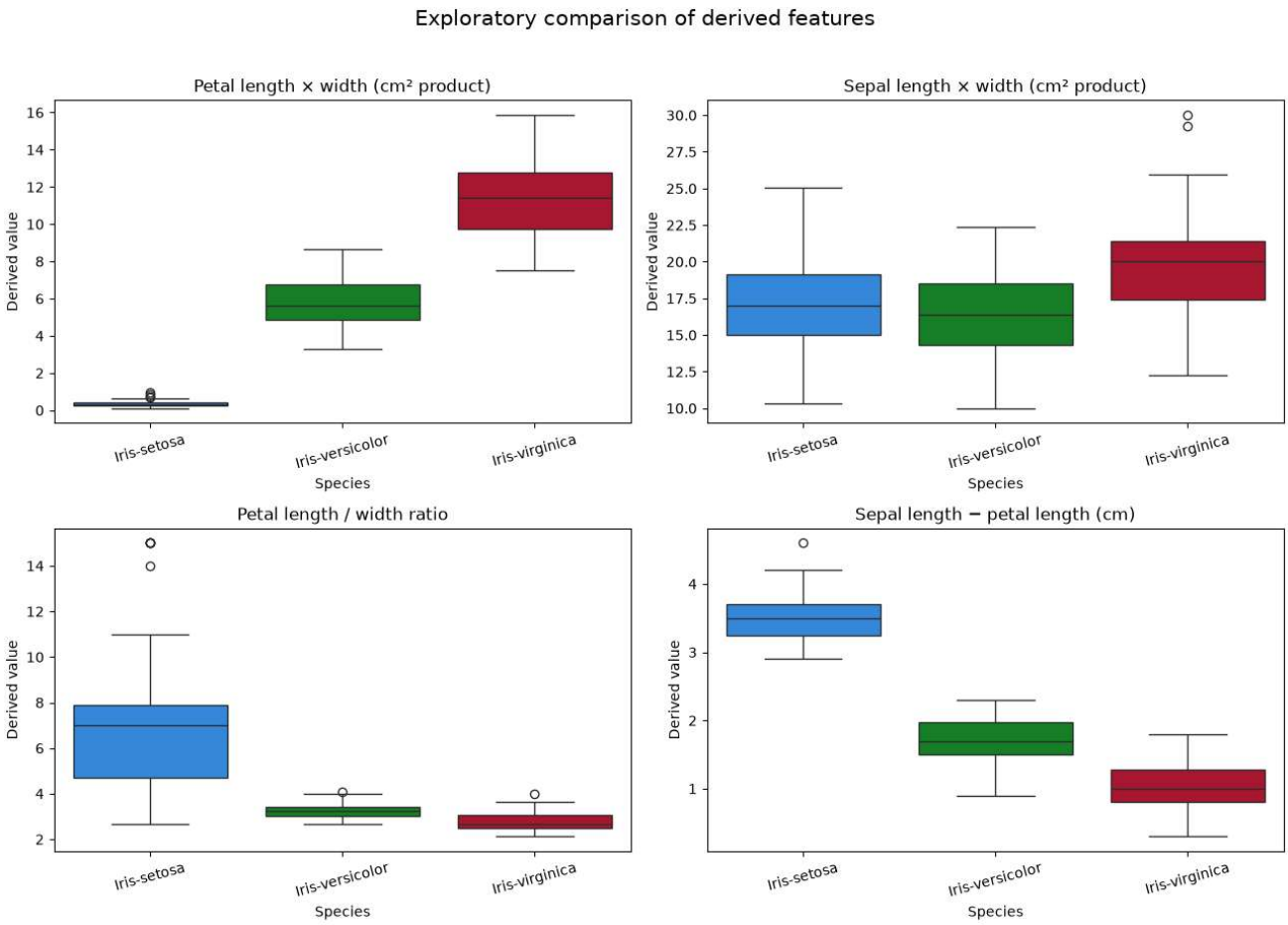
5. Exploring derived measurements

We calculate four additional values from the original measurements:

- **Petal length × petal width** — combines the two petal dimensions into one value.
- **Sepal length × sepal width** — combines the two sepal dimensions.
- **Petal length ÷ petal width** — compares the relative dimensions of the petal.
- **Sepal length − petal length** — compares the length of the sepal with the length of the petal.

These simple calculations provide another way to explore the data. The two products have units of cm^2 mathematically, but they are not direct measurements of actual surface area.

The purpose is to see whether these derived values make differences between species easier to notice. This analysis does not test whether they improve species identification.



What do the derived measurements reveal?

The product of petal length and petal width makes the differences between *Iris-setosa* and the other two species particularly visible. The values for *Iris-versicolor* and *Iris-virginica* still overlap.

These calculated values present the original measurements in a different way; they do not add new measurements to the dataset.

The charts suggest that the petal-length-by-width product may be useful for describing differences between species. However, this visual analysis does not establish whether it improves species identification.

6. Relationships between measurements

What should we look for?

These charts show which measurements tend to change together and whether the patterns are similar across all flowers or differ between species.

In the heatmaps, red shades indicate positive relationships, while blue shades indicate negative relationships. Stronger colours indicate clearer relationships.

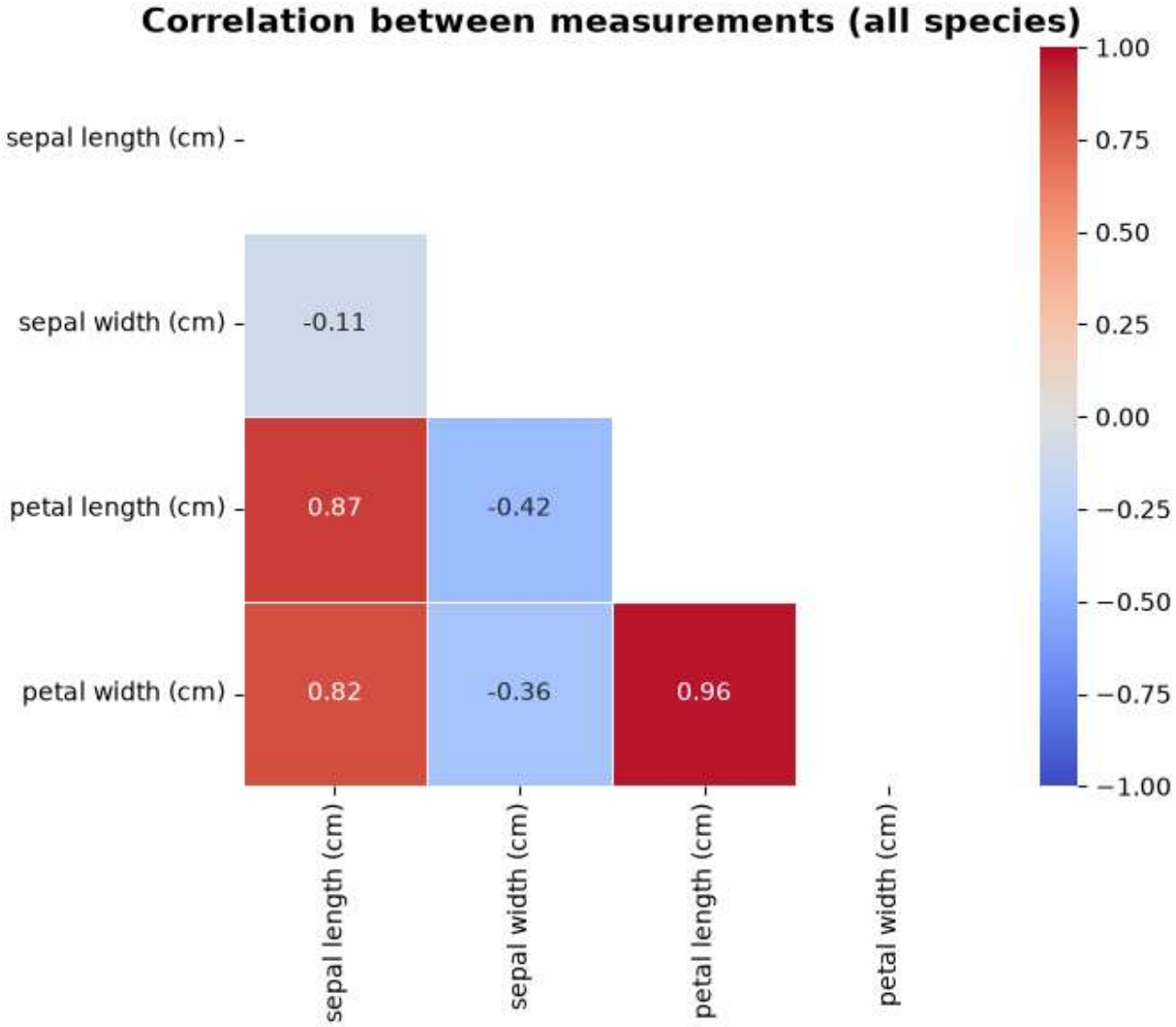
Only half of each matrix is shown because the other half repeats the same information.

Start with the chart for all species, then compare it with the charts for each species separately.

Looking at all species together

The clearest relationship is between petal length and petal width: flowers with longer petals generally also have wider petals.

Because all three species are shown together, this pattern may partly reflect differences in their typical petal sizes. The next charts help us see whether similar relationships appear within each species.

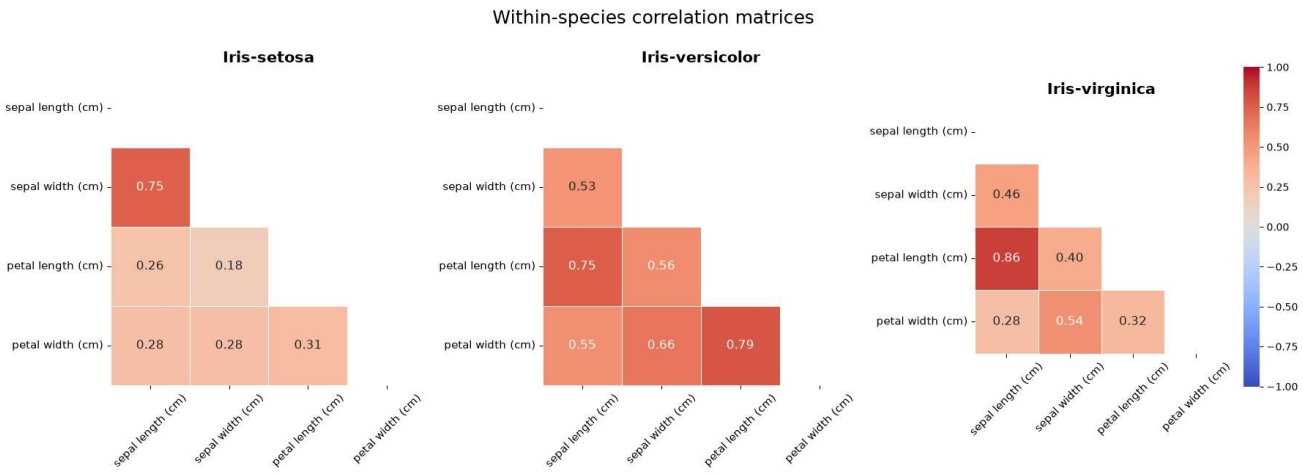


How do the patterns differ by species?

The charts show that the relationships between measurements are not identical for all three species.

- **Iris-setosa:** sepal length and sepal width show the clearest relationship.
- **Iris-versicolor:** petal length and petal width stand out.
- **Iris-virginica:** sepal length and petal length show the clearest relationship.

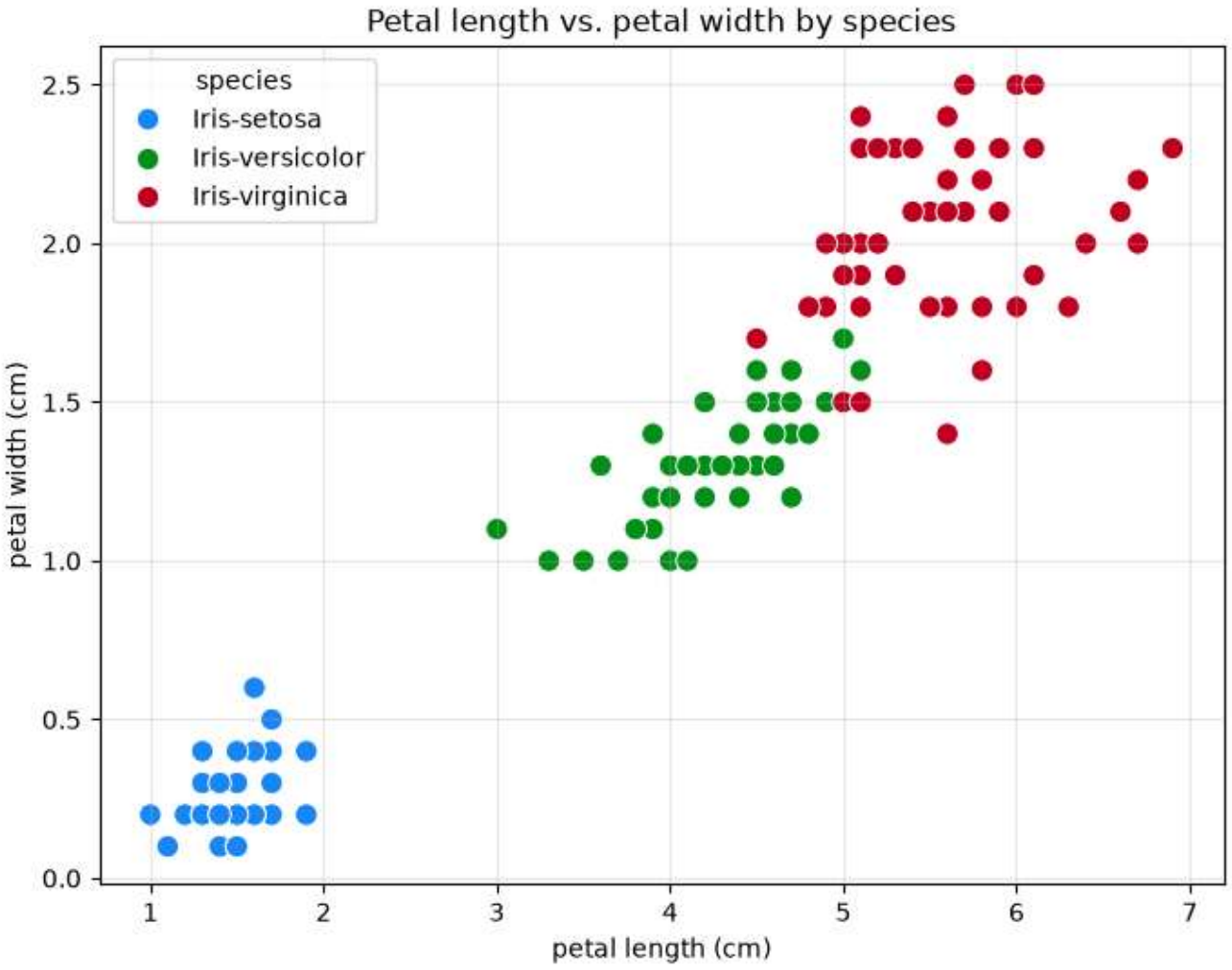
This comparison reveals patterns that may be less obvious when all flowers are examined together.



Petal length versus petal width

This chart compares the two petal measurements for every flower.

Focus on how clearly *Iris-setosa* forms a separate group and where *Iris-versicolor* and *Iris-virginica* overlap.



What does the chart show?

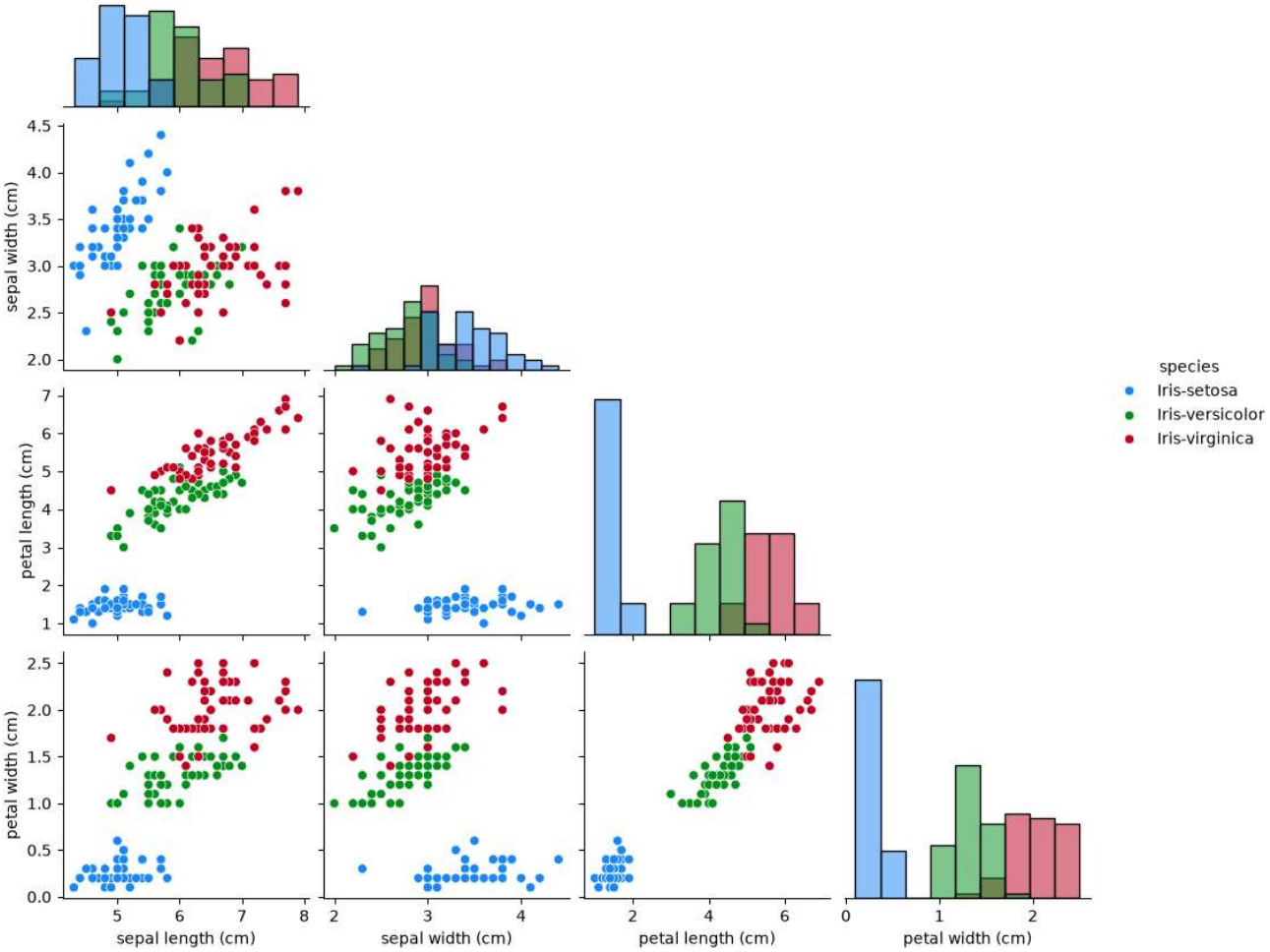
The points for *Iris-setosa* form a clearly separated group. *Iris-versicolor* and *Iris-virginica* occupy nearby regions and overlap in part.

This means that petal measurements show clear visual differences between the species, particularly for *Iris-setosa*. However, the chart does not establish whether every new flower could be identified correctly.

Comparing all measurement combinations

This overview brings together the distributions of all four original measurements and shows how pairs of measurements relate to one another.

Look for combinations that form clearly separated groups and for those where the species overlap.



What stands out across the charts?

The clearest separation appears in combinations involving petal length and petal width. Sepal measurements show more overlap between species.

This supports the observations from the histograms and the petal comparison chart: petal measurements provide the clearest visual differences between the species in this dataset.

7. Findings and limitations

Main findings

- The dataset contains **150 flowers**, with 50 records for each of the three species.
- No missing values were found. Three rows are exact duplicates; they have been retained because identical measurements alone do not prove that a record is incorrect.
- Petal length and petal width show the clearest visual differences between the species, particularly for *Iris-setosa*.
- *Iris-versicolor* and *Iris-virginica* overlap in some measurement ranges, so the charts do not show a perfect separation between these two species.
- The relationships between measurements differ depending on whether all species are viewed together or each species is examined separately.
- The additional calculated measurements provide another perspective on the data, but their effect on species identification has not been tested.

Limitations

- The dataset is relatively small and may not represent the conditions encountered when measuring flowers in a real-world study.
- The findings are based on visual comparisons. We have not tested how reliably the measurements could identify new flowers.
- The product of petal length and petal width is a calculated indicator, not a direct measurement of the petal's actual surface area.
- Unusual measurements and identical records have not been removed automatically. Further evidence would be needed before deciding whether any observations should be excluded.

Conclusion

Petal measurements show the clearest visual differences between the three Iris species. *Iris-setosa* forms the most distinct group, while *Iris-versicolor* and *Iris-virginica* overlap in some regions.

The visualisations help us understand the dataset and identify which measurements appear most useful for distinguishing the species. They do not, however, establish how reliably these measurements would identify new flowers.